LAB University of Applied Sciences

Task-V

WEB GUI testing

LAB University of Applied Sciences
Bachelor of Engineering: Industrial Information Technology
2027
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Statement about using AI

The author of this thesis, Davin Pihaljasaari, used the following AI tools during the preparation of the thesis:

ChatGPT

Purpose of use: used to make the WEB GUI for my converter function.

Explanation of the use of the tool: I had ChatGPT generate a basic GUI that used my JavaScript function and had it add comments after I was done modifying the files.

Responsibility

The author of this thesis, Davin Pihlajasaari, takes full responsibility for the content of this thesis and has reviewed and edited the content generated by the possible use of AI tools. The author has saved all discussions with AI applications during the thesis process. Thus, the extent of the author's own work and the use of AI can be verified if necessary.

The authenticity of this thesis has been checked with the Turnitin similarity checking software.

Abstract

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Task-V

WEB GUI testing

Degree, Field of Study

Bachelor of Engineering: Industrial Information Technology

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Petri Rantanen

Abstract

A report on WEB GUI testing using both manual and automatic means.

Keywords

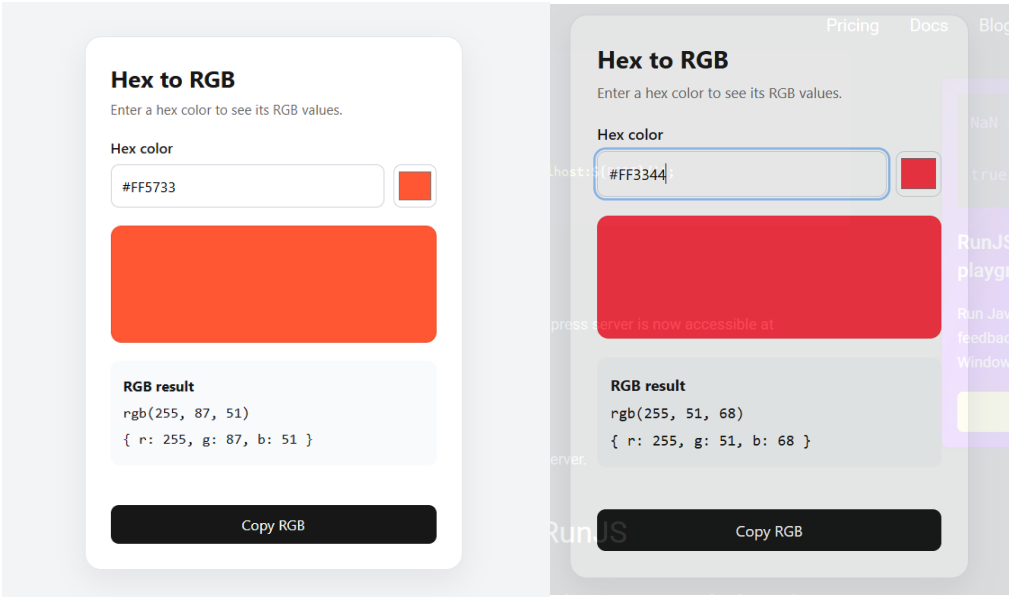
WEB DEV. UI. GUI, testing,

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1 Tests used

The testing methods I used were the same as the previous task; I also did some general input testing manually to make sure that the input validation was working. In general, what I tested for was: did the page load, did the buttons work, was there any overflow and how it handled resizing the window. I chose to redo the automated tests from the last task as there were a couple of tests in it that were good for testing my web app. I did input validation manually because making an automated one is not needed for an application of this simplicity.




```
ui-test-code.py::test_page_loads PASSED

ui-test-code.py::test_page_has_body PASSED
ui-test-code.py::test_page_has_no_horizontal_overflow PASSED
ui-test-code.py::test_images_load PASSED
ui-test-code.py::test_links_have_destinations PASSED
ui-test-code.py::test_buttons_are_interactable PASSED
ui-test-code.py::test_no_broken_internal_links PASSED
ui-test-code.py::test_page_has_no_horizontal_overflow_after_resize PASSED
```

2 Advantages and disadvantages

Using an old automated test would mean that the some of them would not be tailored for this webpage and that means there could be holes in the tests. While using an old, generalized test means that you could check for the most common issues relating to webpage design. Manual testing has limits and most of those limits are because manual testing can’t huge batches of test points. The benefits of manual input testing are that you can just go through a list of the most common inputs that produce errors to see if your system works.

3 Unimplemented alternatives

I could have implemented a rate test by constantly editing the hex code to see if the server and browser could handle the constant input changes. As the system was an automatically updating value converter there were no buttons to click to get the output so there are barely any tests to do on the webpage.

References

Link to GitHub:
https://github.com/DavinPihlajasaari/Devops_engineering/tree/main/Petri/task-V-GUI-testing